

ARUSHI SANJAY AGRAWAL

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SUMMARY

Machine Learning Engineer with 3+ years shipping production ML end-to-end — from research and model development through distributed pipelines to deployment at scale, with prior experience at large technology organisations. A computer vision and 3D perception specialist, currently one of two ML engineers at a 6-person startup, owning its defect-detection and point-cloud pipeline from model architecture to production serving.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript/JavaScript, SQL, R, MATLAB, HTML/CSS, LaTeX

ML / DL Frameworks: PyTorch, detectron2, Mask2Former, Swin Transformer, Ultralytics YOLOv8/YOLO11, TensorFlow, Keras, Scikit-learn, Hugging Face

Computer Vision & 3D: Instance segmentation, RGB-D / multi-modal fusion, custom loss design, auto-labelling / weak supervision, Open3D, RANSAC, point-cloud geometry, OpenCV, model evaluation (mAP / IoU / Dice / F1)

GenAI & LLM: RAG, LoRA fine-tuning, LangChain, FAISS, vector databases, prompt engineering

MLOps & Serving: Ray, FastAPI, Docker, Kubernetes / AKS, Azure, CI/CD, profiling, Prefect, CUDA, ONNX

Data & Tools: NumPy, Pandas, Matplotlib, Git, MySQL

PROFESSIONAL EXPERIENCE

Inframind Labs

Machine Learning Engineer

London, UK

Oct 2025 – Present

- Own the end-to-end computer-vision defect-detection and point-cloud ML pipeline for an automated tunnel-inspection product — a YOLO11 instance-segmentation model detecting 5 masonry-defect classes across 7+ rail and highway tunnel assets, spanning model architecture, distributed inference, and production API serving.
- Built a 4-channel RGB-D Swin Transformer backbone for Mask2Former (detectron2) by inflating pretrained 3-channel patch-embedding weights to ingest depth, and authored a custom recall-aware segmentation loss in an Ultralytics YOLO fork — improving 3D defect recall by 27.6% on the depth+ROI variant.
- Built a 3D→2D auto-labelling pipeline projecting 3D-annotated defect point clouds onto rendered 2D views (stable point-ID overlap, KD-tree matching), parallelised across a Ray cluster to auto-generate YOLO segmentation labels across 6+ tunnels and eliminate manual 2D annotation.
- Designed a view-order-independent 3D evaluation harness (union-find deduplication on mask IoU; Precision/Recall/F1, Dice, IoU, recall-at-coverage) and used it to diagnose model over-segmentation — purity ≈ 0.53 at coverage ≈ 0.94 — as an inference-time fix rather than a costly retraining problem.
- Built a multi-scale detection-fusion algorithm combining predictions across rendered views via noisy-OR point-probability fusion, mask-IoU grouping, and 3D-centroid union-find clustering, eliminating duplicate and fragmented detections.
- Led migration of an 8-workflow point-cloud ML pipeline from Prefect+Ray to pure Ray, authoring a reusable 6-phase playbook and cutting end-to-end runtime by 8–88% through profiler-driven CPU tuning — flagship workflow 121→14 min (–88%, 8.5×), lifting parallelism 0.97×→13.5×.
- Designed and shipped a FastAPI service (submit / probe / result endpoints) proxying jobs to a Ray cluster via the Ray Jobs API with typed Pydantic contracts, containerised and deployed to Azure Kubernetes Service (AKS) with per-asset cluster routing and HPA autoscaling.

Huawei

Reinforcement Learning Intern

London, UK

Apr 2025 – Sep 2025

- Developed a foundation model for scene graph generation from robotics video data, leveraging DINO, CLIP, SAM, and TAM for object detection, segmentation, and tracking.
- Fine-tuned Vision-Language Models using LoRA, improving spatial reasoning and action grounding.
- Integrated graph-based spatial representations into VLA architectures, achieving an 8% improvement on internal contextual reasoning benchmarks.
- Conducted robustness and generalisation analysis, demonstrating overfitting in existing VLA models and identifying key limitations in cross-environment performance.

Adobe

Bangalore, India

Software Development Engineer (GenAI Team)

Jul 2022 – Aug 2024

- Engineered multimodal ML features and integrated them into Adobe Premiere Pro (30M+ users), using C++ and an MVVM architecture to streamline video-editing workflows and increase user satisfaction.
- Optimised a SegNeXt-based model for video mask generation using ONNX, improving accuracy by 17%.
- Analysed Hadoop-sourced user interaction data, driving a 13% increase in feature adoption through data-driven insights.

Text Mercato Solutions

Remote

Machine Learning Intern

Sep 2021 – Mar 2022

- Implemented U2-Net for image segmentation and Boundary-Aware PoolNet for salient object detection, enabling automated product-description generation with NER and LSTMs, reducing inference latency by 50%.

Adobe

Remote

Research Intern

May 2021 – Jul 2021

- Processed 100M+ click-impression logs from Adobe Stock to analyse trends in multimodal asset platforms; deployed with Flask.
- Filed a patent for a novel User Behaviour Click Model using EM and time-series forecasting to predict user affinity scores.

PROJECTS

Retrieval-Augmented Language Agents for Game Play

Jan 2025 – Apr 2025

Open-Endedness and AGI / Prof. Tim Rocktäschel, UCL [\[GitHub\]](#), [\[Paper\]](#)

- Built an LLM-based agent using RAG and a FAISS vector database for long-horizon game environments.
- Trained a reward-ranking model with RLHF, integrating a decision transformer for long-term action evaluation.

Memory-Augmented Critique for Self-Improvement in LLMs

Jan 2025 – Apr 2025

Statistical Natural Language Processing / Prof. Pontus Stenetorp, UCL [\[GitHub\]](#), [\[Paper\]](#)

- Developed a multi-agent system with Super Tiny Language Models (77M parameters) using LoRA, prompt engineering, structured debate, and memory modules, achieving a ROUGE-1 F1 of 0.406 on dialogue summarisation.

EDUCATION

University College London (UCL)

London, UK

MSc Machine Learning — Distinction

2024 – 2025

- Thesis: Enhancing Spatial Reasoning in VLA Models (in collaboration with Huawei).
- Coursework: Probabilistic & Unsupervised Learning, Bayesian Deep Learning, Applied ML, Applied DL, Statistical NLP, Computer Vision, Open-Endedness & General Intelligence.

Indian Institute of Technology Guwahati (IIT-G)

Guwahati, India

B.Tech Engineering Physics — GPA 8.45/10

2018 – 2022

- Thesis: Handling Bias in Visual Question Answering.
- Relevant coursework: Pattern Recognition & Machine Learning, Linear Algebra, Probability & Statistical Mechanics, Computing, Computational Physics.

LEADERSHIP & ACHIEVEMENTS

- **Advanced C++ Boot Camp, Adobe:** Ranked 1st by designing and implementing a high-performance application.
- **Shadowfax Cascade Cup 2022:** 1st place out of 1,086 participants in a national Data Science quiz.
- **JEE 2018:** 99.5 percentile among 1.26 million candidates.
- **Academic Mentor (2021–2022):** Guided a team of 6 students in academics and career planning.